Explainable AI in Radiology

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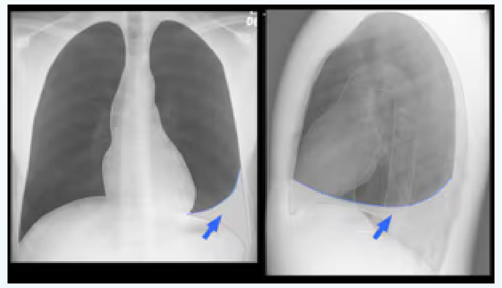
Abstract

Chest X-rays are a common diagnostic tool for lung diseases, including Pleural Effusion.
Explainability is essential in medical imaging AI to gain clinician trust and provide insight into model decisions.

Goal: Train a deep learning model to detect Pleural Effusion and generate explainable visual and example-based explanations to assist radiologists.



Methodology

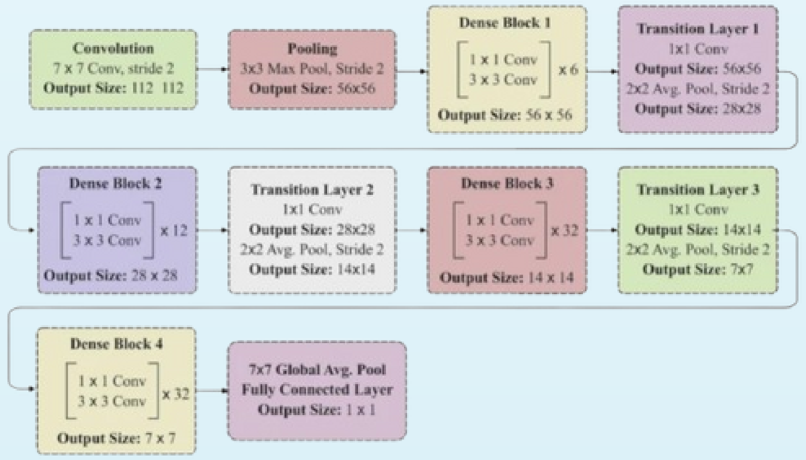
A downsized version of the CheXpert dataset was used, containing both frontal and lateral chest X-rays.

Preprocessing pipeline:

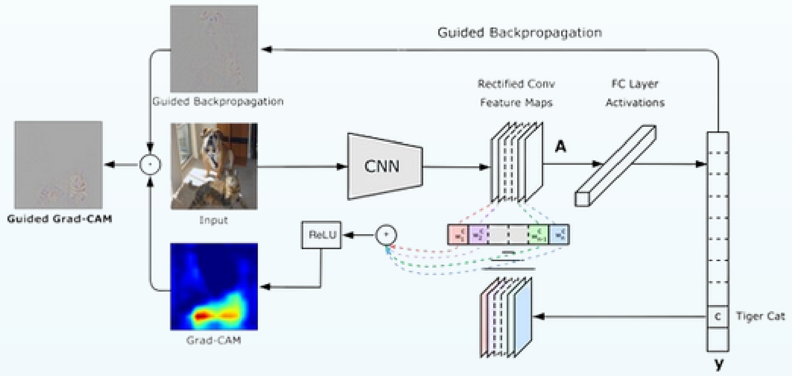
- **Dataset:** Frontal X-rays only, Pleural Effusion labels.
- **Image Prep:** Resized to 224x224, pixels normalized to [0,1].
- **Label Cleanup:** Removed ambiguous/missing labels.
- **Class Imbalance:** Handled using class weights.
- **Train/Validation Split:** 80% train / 20% validation.
- **Channels:** Duplicated to 3 channels for RGB compatibility.

Methods

- **Densenet 169**:
  - Total of 169 layers: Consists of four Dense blocks with Transition, Convolution and Classification Layers.



- **Grad-CAM (Gradient-weighted Class Activation Mapping):**
  - Grad-CAM produces saliency maps that uses "alpha values" which highlight the areas the model focuses on when detecting Pleural Effusion.

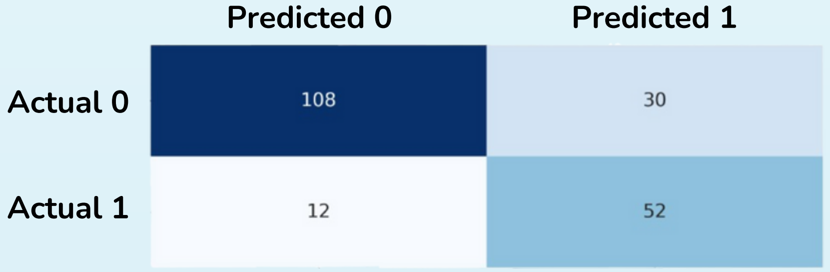


- **Quantus**
  - The Quantus library is used to assess the quality of the AI-generated saliency maps. The evaluation is based on five key metrics: Faithfulness, Robustness, Randomisation, Localisation, and Complexity.
- **Example Based Similarity**
  - The AI model ranks the images using Cosine Similarity between the extracted features. Then the ranked list is compared with an experts ranking. NDCG metric is used to evaluate the performance of how similar AI's judgement to human judgement.

Results

The classification model achieved **high accuracy** with the validation set during training. However, the performance dropped significantly with the test set of 200 images due to imbalance and scarcity.

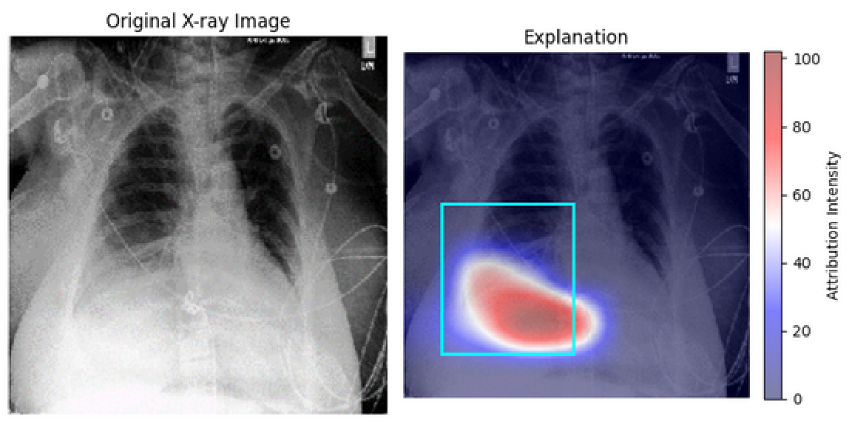
	Accuracy	Precision	Recall
Val Set	0.89	0.87	0,9
Test Set	0.79	0.63	0.81



Confusion Matrix

Saliency Maps

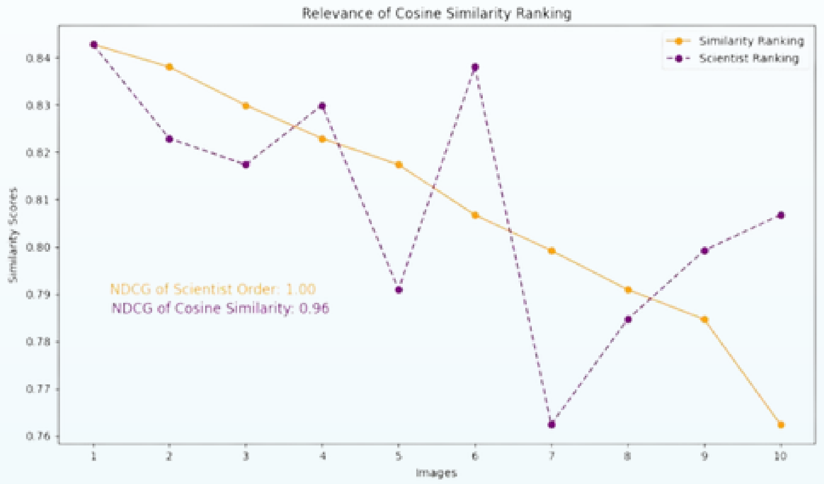
To explain the model, **Grad-CAM** was used.
Grad-CAM produces saliency maps that highlight the areas the model focuses on when detecting Pleural Effusion. It was implemented using the final convolutional layer of DenseNet169, overlaying the heatmaps on the original X-rays.
The saliency maps show that the model consistently highlights the **lower lung regions**, aligning well with radiologists' bounding boxes for Pleural Effusion. Occasionally, the model also highlights unrelated structures, like ribs or medical devices, indicating possible reliance on spurious correlations.



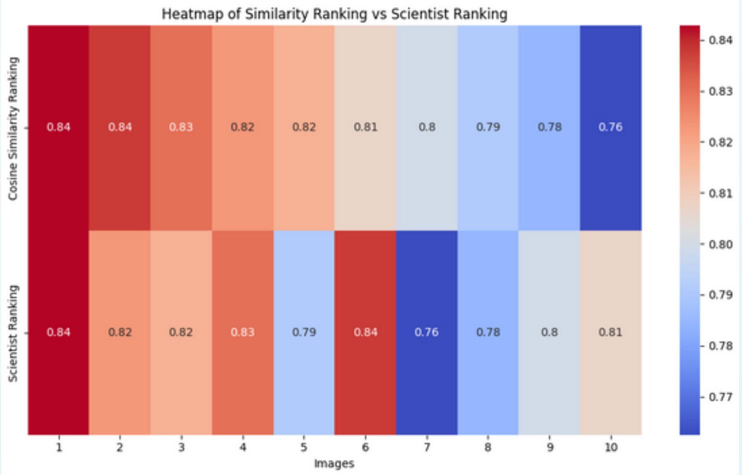
To evaluate the quality of the model's explanations, **Quantus** was used. Quantus provides a range of metrics designed to evaluate how well saliency maps outputs align with desired properties. After evaluating the model it obtained the following results:

- **Robustness** = 0.02. This reflects how stable the model's explanation is under small perturbations to the input. This score indicates moderate robustness, showing that the model tend to stay in similar regions.
- **Complexity** = 9.3. Captures the conciseness of the explanations, in the amount of features used to explain a model's prediction. The score is close to the upper bound based on the number of features used, thus the model balances the use of features for predictions.
- **Randomisation** = Tests to what extent explanations deteriorate as the data labels or the model.
- **Faithfulness** = Checks whether the more important features affect model decisions more strongly.
- **Localisation** = Checks if the explainable evidence is centred around a region of interest.

Example Based Similarity



The plot above shows the ranking of ten images based on opinions of scientists and **cosine similarity,** which is computed with the weights extracted from the last pooling layer of the model.
NDCG score is used for evaluating the relevancy of the ranking based on cosine similarity. The comparison of the ranking showed a good relation between the opinions of the scientists and the features of the model. The heatmap below assigns colors to the images based on their cosine similarity and shows the ranking of the scientists.



Conclusion

The explainability of deep neural network models is crucial for their effective usage in medicine.
The evaluation of saliency maps and the features of the models with the opinions of field experts provide deep insight into the decision-making process of neural network models. The results showed a correlation between the models and the scientist both in critical regions of the images and the weights from the layers of the model.